

A Hybrid Deployment Framework for Industrial Quality Control Using GelSight Tactile Sensing

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Abstract

Tactile sensing offers a potential alternative to vision-based defect detection for reflective, glossy, or low-texture materials common in precision manufacturing. This paper addresses the binary classification of surface defects using GelSight tactile imagery, which captures high-resolution geometric deformations through an elastomeric interface. The primary challenge is the significant “domain gap” between natural images used for pre-training deep learning models and the specific tactile patterns of GelSight sensors, which is further exacerbated by the scarcity of large-scale tactile datasets. To address this, we compare a classical Histogram of Oriented Gradients (HOG) and Support Vector Machine (SVM) pipeline against three ImageNet-pretrained convolutional neural networks (CNNs), including MobileNet V2, ResNet-18 and EfficientNet-B0. The results indicate that in data-constrained tactile perception, hand-engineered features continue to offer unmatched precision and computational efficiency. We recommend a layered deployment strategy for robotic quality control: utilising HOG-SVM for real-time, precision-driven filtering to minimise false positives, and MobileNet V2 for safety-critical scenarios where maximising defect capture is paramount.

1 Introduction

Tactile sensing is a solid solution in scenarios where visual cues alone are insufficient to capture fine-grained geometric details. A new type of tactile sensor, GelSight, provides high-resolution tactile images by measuring surface deformation through an elastomer interface, enabling the detection of defects as small as 1–2 microns, including scratches, dents, and irregularities [Yuan et al., 2017]. GelSight has emerged as a powerful modality for surface inspection in robotics and is well suited to industrial quality control tasks [Yuan et al., 2017].

Extensive research has established the GelSight sensor as a leading modality for high-resolution tactile perception, particularly in fields requiring detailed surface geometry and contact mechanics. For example, GelSight sensors have been used for accurate classification of 13 materials captured from 42 distinct indoor and outdoor items [Shuvo et al., 2026]. Additionally GelSight has been widely utilised in robotics for complex tasks such as dexterous manipulation, slip monitoring, and object grasping, where it enables real-time 3D geometry reconstruction and force estimation through photometric stereo techniques [Cui et al., 2023]. In industrial metrology, tactile sensing can be used for measuring micro-scale features like roughness, scratch depth, and hole diameters on diverse materials, including transparent acrylics where vision-based methods typically struggle. Recent GelSight developments include the “digital fingertip”, a ruggedised, miniaturised version designed for defense robotics to provide real-time tactile intelligence for autonomous systems [Metrology News, 2026]. Furthermore, research in biomedical engineering has introduced specialised variations like TacScope, which can detect subsurface tissue abnormalities up to 5 mm deep, demonstrating the sensor’s versatility in surgical applications [Prince et al., 2025].

In this work, we address the binary classification of defect versus no defect regions using GelSight tactile images of painted manufactured wood products. The defects considered include localised lumps, subsurface adhesive failures beneath the finish, and uneven or degraded paint layers, all appearing at varying spatial locations across the panel and presenting subtle, low contrast tactile signatures. The objective of this study is to compare a

strong classical baseline based on Histogram of Oriented Gradients (HOG) features with a Support Vector Machine (SVM), against a range of ImageNet pretrained convolutional neural networks. While deep learning methods have achieved strong performance in many vision-based tasks [Krizhevsky et al., 2012, Howard et al., 2017], their effectiveness in tactile domains remains uncertain due to differences in data characteristics and reliance on large scale pre-training [Raghu et al., 2019]. The comparison focuses not only on standard performance metrics such as F1 score, precision, recall, and Area over the curve (AUC), but also on robustness to perturbations and practical deployability in resource constrained industrial settings. Through this analysis, we aim to identify the most suitable approach for tactile defect detection under realistic data and operational constraints.

2 Methodology

2.1 Dataset and Preprocessing

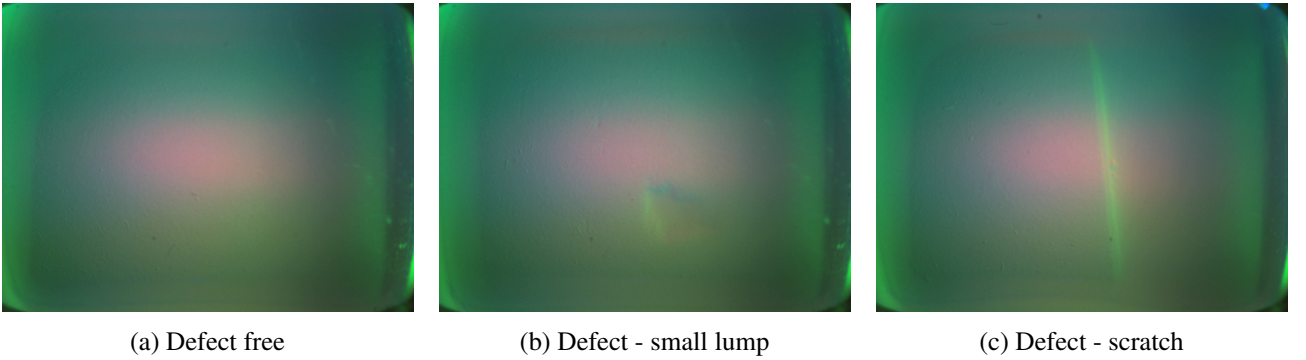


Figure 1: GelSight tactile imagery samples acquired from painted manufactured wood panels under controlled robotic scanning conditions: (a) a defect-free surface baseline; (b) a localised subsurface micro-lump defect presenting subtle geometric deformation; and (c) a macro-scale surface scratch showing sharp directional gradient changes

The tactile image dataset (samples of which are presented in Figure 1) was collected using a robotic manipulator (Universal Robots UR5e) to apply motion parameters across both defect and non-defect regions on painted manufactured wood products using the GelSight elastomeric sensor [Adelson et al., 2011]. This controlled setup ensures reproducible acquisition of images by creating reliable surface scanning conditions critical for reliable model evaluation. The motion parameters were fixed as: 20 mm/s velocity, 0.2 m/s² acceleration, and -0.25N tool contact force [Universal Robots, 2023]. This controlled acquisition ensures reproducible sensor-surface interaction across all 320 high-resolution 860×860 pixel images approximating robotic industrial inspection conditions [Xie et al., 2021]. The dataset consists of 192 defect and 128 non-defect images stratified into a conventional 80/10/10 train-validation-test split using a fixed random seed, preserving an exact 60:40 defect-to-no-defect ratio across all subsets [Pedregosa et al., 2011]. This eliminates stochastic partitioning variation, enabling fair comparison between classical learning and deep learning approaches while reflecting quality assurance sampling where defects constitute 30–50% of inspected batches.

For Deep learning approaches, images are resized, Contrast Limited Adaptive Histogram Equalization (CLAHE) enhancement applied [Shorten and Khoshgoftaar, 2019], with random horizontal flips and ± 0.05 rad rotations for pose robustness. Data augmentation is applied on-the-fly only during training. We utilise the `RandomFlip` layer configured for horizontal shifting alongside `RandomRotation(factor=0.05)`. Here, the factor is interpreted as a fraction of a full 2π rotation, which results in $\pm 18^\circ$ of geometric rotation [Chollet, 2017]. These augmentations introduced controlled pose variation consistent with industrial robot repeatability constraints and improved robustness to boundary artifacts introduced by padding, without distorting defect textures [Shorten and Khoshgoftaar, 2019]. Classical methods used the original feature representations

without augmentation, but were trained and tested on the same splits, ensuring that performance differences come from the models themselves rather than differences in data exposure.

2.2 Classical Baseline: HOG-SVM

In order to process images using Histogram of Oriented Gradients (HOG) feature extraction, the images were first resized to 128×128 pixels then converted to greyscale [scikit-image Developers, 2020]. This balances detail preservation with computational efficiency. HOG features were extracted using 9 orientations covering 0° – 180° , with 8×8 pixel cells grouped into 2×2 blocks (50% overlap) and L2-Hys normalisation [Dalal and Triggs, 2005]. Feature vectors of 8100 dimensions were generated which effectively captured key indicators for surface defects including: the local gradients, edge directions, and contact patterns in the tactile images [Dalal and Triggs, 2005]. This preprocessing step standardises feature scales, ensuring equitable contributions to the decision boundary and prevents any single dimension from dominating due to magnitude disparities [Pedregosa et al., 2011]. A linear SVM, implemented within scikit-learn’s `StandardScaler-SVM` pipeline [Pedregosa et al., 2011], using the default decision threshold of $t = 0.5$, was trained to classify the HOG features. The hyperparameters of the model were optimised by 5-fold stratified `GridSearchCV` on $C \in \{0.1, 1, 10\}$ with `class_weight = balanced` to maintain the inherent 60:40 defect: no-defect imbalance, selecting $C = 0.1$ which maximised cross-validation F1-score whilst preventing overfitting on the 320-image dataset [Pedregosa et al., 2011, Ghosh et al., 2019]. HOG-SVM is well known to excel on small, structured datasets, enabling visualisation of the model’s learned feature weights to identify the edge orientations most critical for defect discrimination [Dalal and Triggs, 2005]. This interpretable classical pipeline establishes a robust baseline for evaluating deep learning approaches in data-constrained industrial tactile inspection scenarios [Chang and Lin, 2011].

2.3 Deep Learning Models

MobileNetV2 [Sandler et al., 2018], ResNet-18 [He et al., 2016], and EfficientNet-B0 [Tan and Le, 2019] were selected as the pre-trained architectures used to classify the GelSight tactile images. Together, they cover a range of model complexities, from lightweight networks suitable for edge deployment to more computationally intensive models that serve as accuracy references. All models were trained on identical 224×224 CLAHE-preprocessed images [Pizer et al., 1990], with a stratified 60:40 defect to non-defect ratio preserved across the train, validation, and test splits. This ensures that performance differences reflect model capacity [Pedregosa et al., 2011]. In order to allow models to retain useful low-level tactile features, while adapting the higher-level layers to specific patterns, certain parameters were selected. Training was controlled using early stopping based on validation AUC (patience = 15 epochs, minimum delta = 0.001) [Prechelt, 1998]. In terms of fine-tuning, the final 20 layers of each network were updated using the Adam optimiser [Kingma and Ba, 2014] with a learning rate of 1×10^{-5} .

Threshold optimisation addressed classification boundary sensitivity in MobileNetV2, where the default 0.5 decision boundary yielded suboptimal F1-scores due to tactile image variability. F1-optimal thresholds were determined via precision-recall curve analysis on the validation set, selecting the threshold that maximised the F1-score for imbalanced defect detection [Davis and Goadrich, 2006]. To ensure robustness, candidate thresholds identified from the validation set were further evaluated using stratified K -fold cross-validation [Pedregosa et al., 2011]. These candidates included Youden’s index ($t = 0.835$) [Youden, 1950] and a recall-constrained threshold ($t = 0.615$). Cross-validation was then used to evaluate intermediate operating regions, resulting in a final fixed threshold of $t = 0.73$ for all subsequent testing and robustness experiments. This final threshold provides a balanced trade-off between precision and recall under tactile image variability.

3 Results

We conducted performance evaluation of the various machine learning methods using the test set of 19 defects and 13 non-defects. The results are presented in Table 1 and it highlights distinct strengths of different models for

industrial deployment scenarios. HOG-SVM achieved the strongest overall performance with perfect precision and superior F1-score compared with other methods, demonstrating unmatched data efficiency for low-data tactile inspection. This shows that the classical method, though only using a small number of effective parameters, performs well, thus emphasising the effectiveness of gradient-based feature representations in low-data tactile inspection environments. On the other hand, MobileNetV2 achieves perfect recall with a balanced precision. This establishes that the model positively ensures that all defect instances are detected, making it particularly suitable for industrial inspection scenarios with high value products.

Table 1: Performance benchmarking using the GelSight test set; models are ranked by their F1-score

Method	F1	Recall	Precision	AUC	Params
HOG-SVM	0.919	0.850	1.000	0.986	378
MobileNetV2 (A5)	0.790	1.000	0.667	0.914	3.5M
ResNet-18	0.769	0.714	0.833	0.896	11.2M
EfficientNet-B0	0.333	0.214	0.750	0.725	4.1M

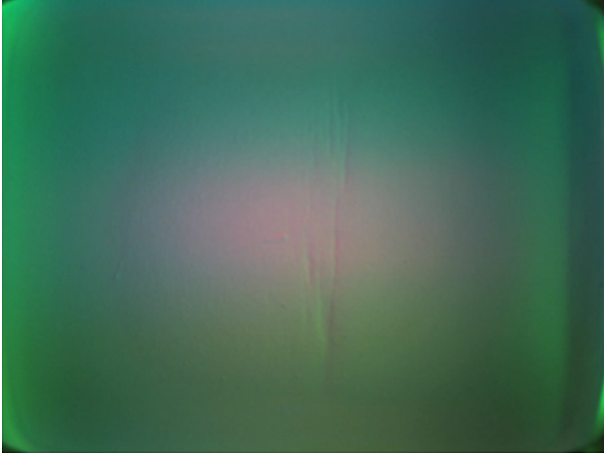
ResNet-18 is potentially a suitable choice for quality control scenarios due to its apparent balanced and consistent performance, whilst EfficientNet-B0 demonstrates moderate capability, suggesting potential use in pre-screening roles, though it did not match the stability of other models. All the CNNs were evaluated at the optimised threshold ($t = 0.73$) to standardise the decision boundary across all deep learning models while HOG-SVM used the default $t = 0.5$ [Youden, 1950]. Additionally, robustness analysis examined the effects of additive noise, blur, brightness variation, and rotation perturbations on the models. The performance of the HOG-SVM pipeline degraded significantly under noise and rotation perturbations. On the other hand, brightness variation did not affect it. This robustness is due to the fact that gradient based descriptors are sensitive to high frequency and orientation perturbation. In contrast, MobileNetV2’s predicted outputs for all perturbations were very similar at the chosen threshold ($t = 0.73$) suggesting saturation of threshold rather than robustness. This indicates that the model focused on defect recall at the expense of sensitivity to input variations, with stable model predictions but low discriminative adaptation under perturbation.

4 Discussion

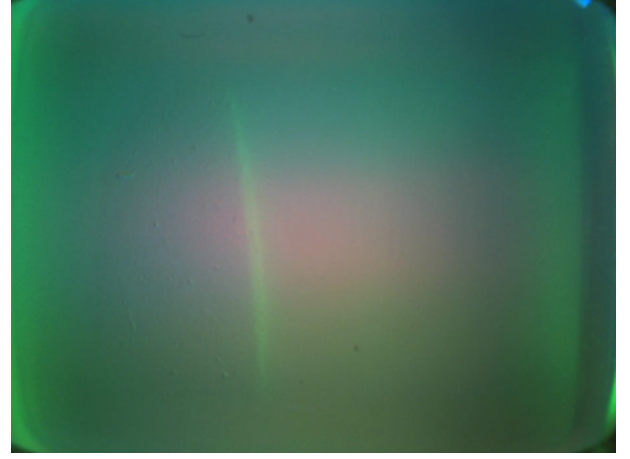
The observed performance differences indicate a clear domain gap between tactile image data and the natural image distributions used for CNN pre-training [Yuan et al., 2017]. Consistent with this mismatch, HOG-SVM achieved a higher F1 score (0.919) than MobileNetV2 (0.790), ResNet-18 (0.769), and EfficientNet-B0 (0.333), indicating stronger adaptation to tactile features. Although HOG-SVM and the CNNs use the same dataset and Keras augmentation, CNNs retain ImageNet pre-training bias toward semantic/texture features [Deng et al., 2009]. In contrast, HOG-SVM extracts gradient-based features directly from GelSight’s geometric deformations [Dalal and Triggs, 2005, Adelson et al., 2011]. This domain mismatch is amplified in small datasets, preventing effective adaptation of CNN feature hierarchies [Chollet, 2017].

Model efficiency analysis shows that HOG-SVM provides strong performance with only 378 parameters, while the CNN-based models require millions of parameters, such as MobileNetV2 (3.5M), ResNet-18 (11.2M), and EfficientNet-B0 (4.1M) (see Table 1) [He et al., 2016]. With only a few thousand effective parameters (grid cells $\times$ 9 bins $\times$ L2 normalisation), the performance of data-constrained regimes depends more on feature-signal alignment than representational capacity as adding complexity without data only increases the difficulty of optimisation [Prechelt, 1998]. The lightweight CNN architectures such as EfficientNet-B0 results further reinforce this point. Despite being designed for efficiency, they rely on carefully balanced scaling and training dynamics that assume access to larger, more diverse datasets [Tan and Le, 2019, Howard et al., 2019]. In this setting, they exhibit instability and underfitting, likely due to sensitivity to learning rate selection and limited ability to specialise high-level features for tactile geometry [Kingma and Ba, 2014].

Post-hoc error analysis revealed that false negatives (Fig. 2(a)) arise from edge artefacts and surface patterns



(a) False negative example



(b) False positive example

Figure 2: Primary failure modes observed during validation and test partitions: (a) False Negative (FN) classification due to low-contrast tactile signatures of real defects; and (b) False Positive (FP) where high-frequency edge artifacts or natural wood grain structures mimic true defect gradients

mimicking defect gradients (e.g., lifted wood tabs, non-conforming grain), while false positives (Fig. 2(b)) correspond to faint, low-contrast deformations (e.g., small lumps, uneven paint) that are difficult to distinguish from background variation [Davis and Goadrich, 2006]. Targeted augmentation and data collection focusing on glossy and low-contrast defects may help mitigate these failure modes [Shorten and Khoshgoftaar, 2019].

These findings support the use of hybrid classical and machine learning approaches for low-data tactile inspection, particularly where surface geometry rather than semantic content governs defect discrimination. Failure cases reinforce this observation, showing that edge artefacts and low-contrast defects benefit from explicit gradient sensitivity. Future work should explore hybrid HOG–CNN ensembles to combine robust geometric representations with learned hierarchical features, particularly as larger datasets become available.

5 Conclusion and Recommendations

This research addresses the critical challenge of high-resolution tactile defect detection in industrial settings where visual inspection is often hindered by surface reflectivity or low texture. Using a dataset of GelSight tactile images captured from painted manufactured wood products, the paper compares a classical hand-engineered approach against modern deep learning models. The core problem identified is the "domain gap" between natural images used for pre-training convolutional neural networks (CNNs) and the specific geometric deformations captured by GelSight sensors, a challenge further compounded by the limited availability of large-scale tactile datasets.

The methodology involved a rigorous comparison between a HOG-SVM pipeline and three ImageNet-pretrained CNN architectures: MobileNet V2, ResNet-18, and EfficientNet-B0. The study utilised 320 high-resolution images with a stratified split to ensure a consistent defect to non-defect ratio across training and testing. While the CNNs underwent threshold optimisation and data augmentation to improve robustness, the classical HOG-SVM model focused on extracting gradient-based features directly from the tactile surface deformations.

The outcomes demonstrate a significant trade-off between precision and recall across the tested models, with precision ranging from 0.667 to 1.000 and recall ranging from 0.214 to 1.000. The HOG-SVM model emerged as the most efficient, achieving a perfect precision of 1.000 and the highest F1-score while utilising only a fraction of the parameters required by deep learning architectures. Conversely, MobileNet V2 achieved perfect recall, ensuring that every defect was detected, though at the cost of more false positives. Other lightweight models like EfficientNet-B0 struggled with instability, suggesting that representational capacity is less important

than feature-signal alignment in data-constrained tactile environments.

In conclusion, the paper recommends a layered deployment strategy for industrial robotic inspection. HOG-SVM is proposed as a primary classifier for real-time, precision-driven applications to minimise false alarms, while MobileNet V2 is better suited for recall-critical scenarios followed by human verification. Future work should explore hybrid HOG-CNN ensembles to combine the strengths of geometric feature extraction with hierarchical learned features as larger, more diverse tactile datasets become available for complex robotic quality control [Dalal and Triggs, 2005, Yuan et al., 2017].

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